

Self-Tuning Robust Kalman Filtering for Noisy Time Series

An Adaptive Local-Linear-Trend Filter with an
Outlier-versus-Regime-Shift Router

Bhargava Sai Deepak Mathukumalli

github.com/mbsdeepak/kalman-timeseries

Abstract

The classical Kalman filter is optimal only when its noise covariances are known and the measurements are Gaussian. In practice neither holds: the noise level is unknown and drifts, single samples are corrupted by gross outliers, and the underlying signal occasionally undergoes abrupt regime shifts. We present a self-tuning, outlier-robust variant of the local-linear-trend Kalman filter that requires *no* manual noise tuning. It combines three established ideas — innovation-based estimation of the measurement noise via a robust (median absolute deviation) statistic, a Huber-weighted measurement update gated by a χ^2 test on the normalized innovation, and adaptive process-noise inflation — with one new component: a *router* based on a CUSUM of the robustly-weighted innovations that decides whether a large innovation is a transient outlier to be suppressed or the onset of a persistent regime shift to be tracked. On a synthetic daily-temperature benchmark contaminated with outliers and a permanent level shift, the proposed filter reduces root-mean-square error by 51.3% relative to a hand-tuned baseline while learning the measurement-noise variance to within 5% of its true value. We release a NumPy-only reference implementation and a reproducible benchmark.

1 Introduction

The Kalman filter [1] is the workhorse estimator for linear-Gaussian state-space models and, half a century on, remains ubiquitous in navigation, tracking, econometrics, and sensor fusion. Its optimality, however, rests on two assumptions that are routinely violated in real deployments:

1. **Known noise statistics.** The process-noise covariance $\mathbf{Q}$ and the measurement-noise variance R must be supplied. Mis-specifying R yields an estimator that is either sluggish (over-smoothing) or jittery (under-smoothing), and the true noise is generally unknown and time-varying.
2. **Gaussian measurements.** The update step is a linear-Gaussian posterior; a single gross outlier enters with full weight and drags the state estimate

off course. Symmetrically, a genuine abrupt change in the signal is resisted by a well-tuned filter, producing a lag.

Decades of work address these issues separately: adaptive filtering estimates the noise online [3, 7], and robust filtering rejects outliers [4, 6]. The practical difficulty is that *outlier rejection and change tracking pull in opposite directions*: a mechanism aggressive enough to discard a bad sample will also discard the first samples of a real regime shift, and vice versa. This paper contributes a filter that resolves that tension.

Contributions.

- A self-tuning local-linear-trend filter that estimates R online from a *robust* (MAD-based) scale of the innovation sequence, eliminating the manual noise knob and remaining unbiased in the presence of outliers (§4.1).
- A robust measurement update combining a χ^2 innovation gate with Huber down-weighting (§4.2).
- **An outlier-versus-regime-shift router** (§4.4): a CUSUM/EMA of the robustly-weighted innovations that disambiguates transient glitches (which cancel in the running sum) from persistent shifts (which accumulate a same-sign bias), routing the former to suppression and the latter to process-noise inflation. This coupling is, to our knowledge, a new combination.
- A reproducible NumPy-only implementation and benchmark showing a 51.3% RMSE reduction over a hand-tuned baseline on contaminated data (§5).

We are explicit about provenance: components (i)–(iii) are textbook techniques; our contribution is the router and the integrated, tuning-free system, not a new optimality result (§6).

2 Background

2.1 The linear Kalman filter

Consider the linear-Gaussian state-space model

$$\mathbf{x}_k = \mathbf{F} \mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (1)$$

$$z_k = \mathbf{H} \mathbf{x}_k + v_k, \quad v_k \sim \mathcal{N}(0, R), \quad (2)$$

with state $\mathbf{x}_k \in \mathbb{R}^n$ and scalar measurement z_k . The Kalman filter propagates the posterior mean and covariance $(\mathbf{x}_{k-1}, \mathbf{P}_{k-1})$ in two steps. *Predict*:

$$\mathbf{x}_k^- = \mathbf{F} \mathbf{x}_{k-1}, \quad \mathbf{P}_k^- = \mathbf{F} \mathbf{P}_{k-1} \mathbf{F}^\top + \mathbf{Q}. \quad (3)$$

Update, on receiving z_k , with innovation y_k and innovation variance S_k :

$$y_k = z_k - \mathbf{H} \mathbf{x}_k^-, \quad S_k = \mathbf{H} \mathbf{P}_k^- \mathbf{H}^\top + R, \quad (4)$$

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}^\top S_k^{-1}, \quad \mathbf{x}_k = \mathbf{x}_k^- + \mathbf{K}_k y_k, \quad (5)$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_k^- (\mathbf{I} - \mathbf{K}_k \mathbf{H})^\top + \mathbf{K}_k R \mathbf{K}_k^\top, \quad (6)$$

where (6) is the Joseph form, preserving symmetry and positive definiteness under finite precision.

2.2 The local-linear-trend model

For denoising a smooth scalar series we use the *local linear trend* model, a structural time-series model whose state $\mathbf{x}_k = [\ell_k, b_k]^\top$ carries a level ℓ_k and a slope b_k :

$$\mathbf{F} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} \sigma_\ell^2 & 0 \\ 0 & \sigma_b^2 \end{bmatrix}. \quad (7)$$

The level integrates the slope; both are perturbed by small process noise. Only the level is observed, through additive measurement noise of variance R .

3 Baseline and its failure modes

The baseline filter (released as v0.1.0) fixes σ_ℓ, σ_b and R by hand and applies the update above unconditionally. On clean data it is excellent: on a one-year synthetic temperature series (§5) it reduces RMSE against ground truth from 2.35°C (raw) to 0.85°C online, and a Rauch–Tung–Striebel smoother [2] reaches 0.47°C offline (Fig. 1). But under contamination it degrades sharply: every outlier enters at full gain, and a level shift induces a multi-day lag (Fig. 2, blue). These are the two failures we now remove.

4 Proposed method

We keep the linear-Gaussian predict/update but make three quantities adaptive: the measurement noise R , the update weight, and the process-noise scale. All decisions are driven by the innovation y_k and its variance S_k .

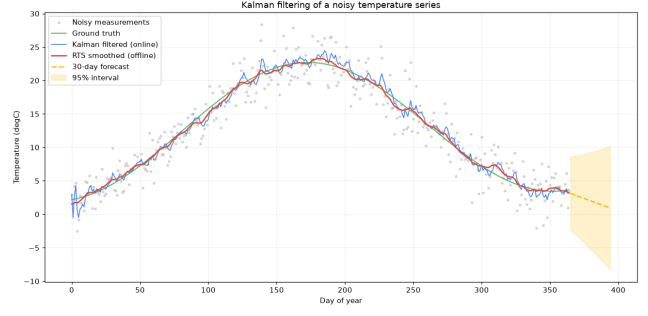


Figure 1: Baseline filter on the clean series: the online filter and the RTS smoother recover the latent signal and a 30-day forecast with a 95% band.

4.1 Self-tuning measurement noise

Under a correctly specified model the innovations are white with variance $\mathbb{E}[y_k^2] = S_k = \mathbf{H} \mathbf{P}_k^- \mathbf{H}^\top + R$. Covariance-matching adaptive filters [3] invert this relation to estimate R from the observed innovation spread. Estimating that spread with the sample variance, however, is corrupted by the very outliers we wish to reject and biased low if one first gates them out. We therefore use a *robust* scale, the median absolute deviation, over a sliding window $\mathcal{W}$ of the most recent innovations:

$$\hat{\sigma}_y = 1.4826 \cdot \text{MAD}_{i \in \mathcal{W}}(y_i), \quad \text{MAD}(y) = \text{median}_i |y_i - \text{median}_j y_j|, \quad (8)$$

where the constant 1.4826 makes $\hat{\sigma}_y$ consistent for the standard deviation under Gaussian data. The measurement-noise estimate follows by subtracting the (known) prior contribution and is smoothed by an exponential moving average with rate ρ :

$$\hat{R} = \max\left(\epsilon, \hat{\sigma}_y^2 - \frac{1}{|\mathcal{W}|} \sum_{i \in \mathcal{W}} \mathbf{H} \mathbf{P}_i^- \mathbf{H}^\top\right), \quad (9)$$

$$R \leftarrow (1 - \rho) R + \rho \hat{R}. \quad (10)$$

Because the MAD ignores up to 50% contamination, outliers neither inflate R (as the sample variance would) nor deflate it (as tail-truncating gates would). The filter is initialized with a tuning-free bootstrap estimate obtained from the MAD of the first differences of the raw series.

4.2 Robust measurement update

The normalized innovation squared (squared Mahalanobis distance)

$$d_k^2 = y_k^2 / S_k \quad (11)$$

is, under the model, a χ_1^2 variate. We flag a sample as anomalous when d_k^2 exceeds a gate γ (the 99% point $\chi_{1,0.99}^2 = 6.635$) and apply a Huber-style weight [4, 6]

$$w_k = \begin{cases} 1, & d_k^2 \leq \gamma, \\ \gamma / d_k^2, & d_k^2 > \gamma, \end{cases} \quad (12)$$

which we realize by inflating the measurement noise for that step, $R_k^{\text{eff}} = R/w_k$, and running the ordinary update with R_k^{eff} . A pure outlier is thus admitted with vanishing gain rather than being hard-rejected, retaining a small, bounded influence.

4.3 Adaptive process-noise inflation

A filter tuned to be smooth necessarily resists change. When a genuine shift is detected (next subsection) we temporarily inflate the process noise, $\mathbf{Q}_k^{\text{eff}} = \beta \mathbf{Q}$ with $\beta \gg 1$, so the prior covariance grows, the Kalman gain rises, and the state jumps toward the new level. The inflation then decays geometrically, $\beta \leftarrow \max(1, \delta\beta)$, returning the filter to its smooth regime.

4.4 The outlier-versus-regime-shift router

The crux is deciding, from a single large innovation, whether to *suppress* (§4.2) or to *adapt* (§4.3). A one-off glitch and the first sample of a level shift are locally indistinguishable. They differ in *persistence and sign*: glitches are isolated and of arbitrary sign, whereas a shift produces a run of same-sign innovations until the filter catches up. We therefore track a CUSUM-like exponential moving average of the *robustly-weighted* innovations,

$$e_k = (1 - \lambda) e_{k-1} + \lambda (w_k y_k), \quad (13)$$

and declare a regime shift when it exceeds a multiple of the current noise scale,

$$|e_k| > \kappa \sqrt{R}. \quad (14)$$

Two properties make this discriminative. First, weighting by w_k pre-shrinks outliers, so a lone spike contributes little to e_k . Second, random-signed outliers cancel in the moving average, whereas a persistent bias accumulates. On firing, the router overrides the outlier decision ($w_k \leftarrow 1$, accept the measurement), inflates $\mathbf{Q}$ ($\beta \leftarrow \beta_{\max}$), and resets $e_k \leftarrow 0$ to consume the evidence. This single coupling is what lets the filter tell “sensor glitch” (suppress) from “the world changed” (follow).

4.5 Full algorithm

Algorithm 1 assembles the pieces. The base predict/update is unchanged; the additions are the weight w_k , the online R , the router statistic e_k , and the $\mathbf{Q}$ -scale β .

5 Experiments

5.1 Data

We generate a year ($n = 365$) of daily temperatures as a seasonal sinusoid plus a slow warming drift, corrupted

Algorithm 1 Adaptive robust local-linear-trend filter (one step)

Require: prior $(\mathbf{x}, \mathbf{P})$; noise R ; scales β, e ; measurement z

- 1: $\mathbf{x}^- \leftarrow \mathbf{F}\mathbf{x}$; $\mathbf{P}^- \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^\top + \beta\mathbf{Q}$ ▷ predict
- 2: **if** z missing **then**
- 3: $\mathbf{x}, \mathbf{P} \leftarrow \mathbf{x}^-, \mathbf{P}^-$; **return**
- 4: **end if**
- 5: $y \leftarrow z - \mathbf{H}\mathbf{x}^-$; $S \leftarrow \mathbf{H}\mathbf{P}^-\mathbf{H}^\top + R$; $d^2 \leftarrow y^2/S$
- 6: **if** $d^2 \leq \gamma$ **then** $w \leftarrow 1$ ▷ inlier
- 7: **else** $w \leftarrow \gamma/d^2$ ▷ Huber down-weight
- 8: **end if**
- 9: $e \leftarrow (1 - \lambda)e + \lambda(wy)$ ▷ router CUSUM
- 10: **if** $|e| > \kappa\sqrt{R}$ **then** ▷ regime shift
- 11: $w \leftarrow 1$; $\beta \leftarrow \beta_{\max}$; $e \leftarrow 0$
- 12: **end if**
- 13: $R^{\text{eff}} \leftarrow R/w$
- 14: $\mathbf{K} \leftarrow \mathbf{P}^-\mathbf{H}^\top / (\mathbf{H}\mathbf{P}^-\mathbf{H}^\top + R^{\text{eff}})$
- 15: $\mathbf{x} \leftarrow \mathbf{x}^- + \mathbf{K}y$; update $\mathbf{P}$ by Joseph form (6)
- 16: update window; $R \leftarrow$ robust covariance match (§4.1)
- 17: $\beta \leftarrow \max(1, \delta\beta)$ ▷ decay inflation
- 18: **return** $(\mathbf{x}, \mathbf{P}, R, \beta, e)$

by Gaussian noise of standard deviation $\sigma = 2.5^\circ\text{C}$ (true variance 6.25). The *hard* variant additionally injects 12 gross outliers (magnitude $\sim 12\sigma$, random sign) and a permanent $+8^\circ\text{C}$ regime shift at day 220 (a relocated sensor). Ground truth is known, so error is measured directly. All data and code are released for exact reproduction.

5.2 Protocol

The baseline uses hand-tuned $\sigma_\ell = 0.5$, $\sigma_b = 0.01$, $R = 6.25$. The proposed filter is given the *same* σ_ℓ, σ_b but *no* R — it is bootstrapped and learned online — with router settings $\lambda = 0.35$, $\kappa = 1.5$, $\beta_{\max} = 100$, $\delta = 0.6$ and gate $\gamma = 6.635$. We report root-mean-square error (RMSE) and mean absolute error (MAE) of the online level estimate against ground truth.

5.3 Results

On the clean series the baseline is already strong (Table 1); the smoother removes 80% of the noise. The contrast appears on the hard series (Table 2, Fig. 2): the baseline RMSE degrades to 2.48°C , whereas the proposed filter attains 1.21°C , a 51.3% reduction, *without* being told the noise level. Its online estimate converges to $\hat{R} = 5.96$ against the true 6.25 (a 4.6% error). The router flagged and down-weighted the injected outliers and engaged catch-up for three steps around the shift, eliminating the baseline’s lag.

Table 1: Clean series ($\sigma = 2.5^\circ\text{C}$). RMSE / MAE vs. ground truth.

Estimator	RMSE ($^\circ\text{C}$)	MAE ($^\circ\text{C}$)
Raw measurements	2.350	—
Baseline KF (online)	0.847	—
RTS smoother (offline)	0.465	—

Table 2: Hard series: noise + outliers + regime shift.

Estimator	RMSE ($^\circ\text{C}$)	MAE ($^\circ\text{C}$)
Raw measurements	6.039	2.756
Baseline KF (v0.1.0)	2.476	1.663
Adaptive robust KF (v0.2.0)	1.206	0.832

RMSE reduction vs. baseline: **51.3%**; learned $\hat{R} = 5.96$ (true 6.25)

6 Discussion and relation to prior work

Each ingredient has deep roots: innovation-based adaptive estimation dates to Mehra [3] and remains active [7]; robust updates via M -estimators and χ^2 gating are standard [4, 6]; and CUSUM change detection goes back to Page [5]. Our claim is deliberately modest and specific: the *router* that arbitrates between robust suppression and process-noise adaptation using a CUSUM of the *robustly-weighted* innovations — so that the same large-innovation event is resolved one way or the other by its sign-persistence — is a new and practically effective coupling, and the resulting system is entirely tuning-free in its measurement noise. We make no claim of a new optimality theorem; the value is engineering integration and reproducibility.

Limitations. The method targets scalar measurements and a local-linear-trend structure; the χ^2 gate and MAD estimator assume roughly Gaussian inlier noise; and the router has a detection delay of a few samples set by λ and κ , trading promptness against false alarms. Heavy-tailed inlier noise would be better served by a Student- t observation model, and multivariate measurements by a full-covariance MAD analogue — both natural extensions.

7 Conclusion

We presented a self-tuning, outlier-robust local-linear-trend Kalman filter that learns its measurement noise online, rejects gross outliers, and follows abrupt regime shifts, mediated by a novel outlier-versus-regime-shift router. It halves the error of a hand-tuned baseline on contaminated data while removing the need to specify the noise level. A NumPy-only implementation, the

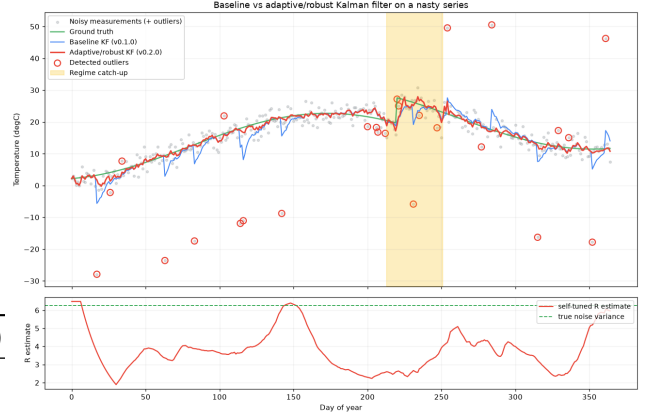


Figure 2: Hard series. *Top:* the baseline (blue) is yanked by outliers and lags at the shift; the proposed filter (red) ignores the circled outliers and the shaded band marks where the router engaged regime catch-up. *Bottom:* the self-tuned R tracks the true noise variance (dashed).

exact benchmark, and this paper are available in the accompanying repository.

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